

MIDLAND BARSTOW, MI, USA

WMO: 720629

Lat: 43.663N

Lon: 84.261W

Elev: 637

StdP: 14.36

Time Zone: -5.00 (NAE)

Period: 09-19

WBAN: 00216

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.1	3.0	-10.2	3.3	-0.5	-4.4	4.4	4.6	20.7	27.1	19.0	25.2	1.9	220	0.514

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.1	89.7	73.1	86.3	70.9	83.9	69.7	75.4	85.1	73.6	82.4	72.1	80.2	8.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.6	123.6	80.0	70.9	116.6	78.2	69.5	111.2	77.0	39.3	85.2	37.6	82.3	36.1	80.0	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 19.4	(b) 17.2	(c) 15.4	(e) -12.2	(f) 93.9	(g) 6.9	(h) 3.3	(i) -17.2	(j) 96.2	(k) -21.2	(l) 98.1	(m) -25.1	(n) 99.9	(o) -30.2	(p) 102.3		
(4) 19.4	(5) 17.2	(6) 15.4	(7) -12.6	(8) 78.8	(9) 6.9	(10) 2.1	(11) -17.6	(12) 80.3	(13) -21.6	(14) 81.6	(15) -25.4	(16) 82.8	(17) -30.4	(18) 84.3		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.5	22.8	24.0	33.4	44.9	58.0	65.8	70.7	68.8	61.8	50.4	38.5	29.6
	DBStd	19.01	11.02	11.22	11.06	8.83	8.86	6.75	6.01	5.44	7.98	8.43	8.80	8.99
	HDD50	3421	844	728	529	199	22	0	0	0	4	102	360	633
	HDD65	6915	1309	1148	980	605	255	70	17	25	155	458	796	1097
	CDD50	2514	0	0	14	46	271	473	640	582	358	115	14	1
	CDD65	533	0	0	0	2	39	92	192	142	60	6	0	0
	CDH74	5571	0	0	9	42	506	1032	2081	1314	527	60	0	0
	CDH80	1640	0	0	0	5	128	302	724	339	137	5	0	0
	WSAvg	5.9	6.7	7.0	6.6	7.6	6.3	5.2	4.4	4.1	4.3	5.7	6.2	6.3
	PrecAvg	28.7	1.5	1.3	2.0	2.8	2.9	3.0	2.4	3.3	3.3	2.4	2.5	1.8
(15) Precipitation	PrecMax	39.5	4.2	3.9	5.7	5.4	7.3	9.1	4.7	12.8	18.3	5.5	6.7	5.2
	PrecMin	18.7	0.2	0.1	0.4	0.9	0.4	0.7	0.3	0.9	0.0	0.6	0.5	0.1
	PrecStd	4.5	0.9	0.9	1.2	1.2	1.5	1.6	1.2	1.9	2.8	1.1	1.4	1.1
	PrecStd	4.5	0.9	0.9	1.2	1.2	1.5	1.6	1.2	1.9	2.8	1.1	1.4	1.1
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.9	57.7	76.8	80.7	88.1	91.1	93.7	90.3	90.4	80.6	68.8	57.6
		MCWB	50.9	51.6	62.6	61.0	69.5	72.7	75.2	72.3	73.0	65.0	57.4	53.7
	2%	DB	45.8	48.4	64.0	72.1	83.8	87.1	90.3	86.3	83.9	74.6	61.6	50.2
		MCWB	41.9	44.0	54.3	57.7	67.5	69.6	74.3	71.5	71.0	63.3	53.4	46.5
	5%	DB	39.8	42.5	56.7	66.1	80.2	83.4	87.1	83.8	80.0	69.6	56.8	45.2
		MCWB	37.3	38.1	49.7	53.8	64.0	69.0	72.0	70.2	68.0	61.3	49.8	42.5
	10%	DB	36.9	38.6	50.1	61.3	75.8	80.3	83.9	81.3	76.6	65.3	52.2	40.9
		MCWB	34.8	34.8	43.6	50.5	62.4	66.7	70.0	68.5	66.2	59.1	47.5	37.9
	0.4%	WB	51.2	52.5	63.4	64.1	72.6	75.9	78.6	75.9	75.5	69.4	59.4	54.9
		MCDB	52.0	55.4	77.9	74.7	84.2	86.8	90.0	83.9	85.1	74.4	64.5	56.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	42.0	44.1	56.8	59.1	69.5	72.8	76.4	73.8	73.0	66.5	54.3	47.2
		MCDB	44.8	48.6	62.2	68.9	78.5	82.9	87.0	81.9	81.3	70.9	60.4	50.0
	5%	WB	38.0	38.4	50.5	55.5	67.4	71.0	74.4	72.4	70.7	62.8	50.7	42.6
		MCDB	40.1	41.7	55.9	64.3	76.1	80.2	83.5	80.5	76.6	68.2	55.7	45.0
	10%	WB	35.1	35.1	43.6	51.5	64.3	68.8	72.5	70.8	68.5	59.2	47.4	38.2
		MCDB	36.9	38.1	50.0	60.3	73.1	76.6	80.3	78.4	74.2	63.9	52.5	40.6
(35) Mean Daily Temperature Range	5% DB	MDBR	14.2	16.1	18.1	20.5	23.0	22.5	23.1	22.2	22.4	19.5	15.9	11.6
		MCDBR	15.1	18.2	27.1	30.2	28.1	25.7	25.6	24.1	24.5	26.1	22.8	16.6
		MCWBR	13.3	14.2	18.2	17.1	13.9	11.9	11.4	11.1	12.4	15.1	15.9	14.0
		MCDBR	14.9	17.6	23.9	24.6	23.6	21.3	21.6	20.4	20.4	21.3	20.9	15.6
	5% WB	MCWBR	13.6	15.0	17.5	16.7	12.9	11.6	11.0	10.4	11.5	15.1	16.3	14.0
		MCWBR	13.6	15.0	17.5	16.7	12.9	11.6	11.0	10.4	11.5	15.1	16.3	14.0
(40) Clear-Sky Solar Irradiance	taub		0.289	0.318	0.328	0.363	0.388	0.417	0.406	0.398	0.375	0.339	0.311	0.282
	taud		2.374	2.286	2.396	2.327	2.310	2.256	2.295	2.312	2.375	2.473	2.493	2.447
	Ebn at Noon		266	276	290	286	280	272	273	272	269	267	258	260
	Edh at Noon		25	32	33	38	40	43	41	38	33	27	22	21
	RadAvg		440	722	1140	1447	1705	1911	1929	1641	1311	816	491	349
(45) All-Sky Solar Radiation	RadStd		41	87	124	125	155	111	97	76	125	85	64	32

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (57 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page